Basic Human Physiology



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

1

Today's Agenda

- **Introduction**
  Definition & organization
- **Digestive System**
  From mouth to large intestine
- **Immune System**
  Barriers · innate immunity · adaptive immunity
- **Other Body Systems**
  Circulatory · Nervous · Musculoskeletal · Endocrine · Respiratory · Urinary · Reproductive
- **Putting It All Together**
  Everyday examples of systems teaming up

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

2

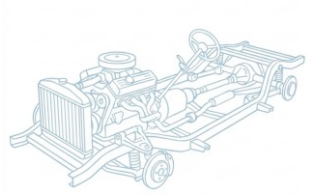
Introduction

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

3

What Is Physiology?

Structure versus **Function**





Anatomy studies the structure of the body

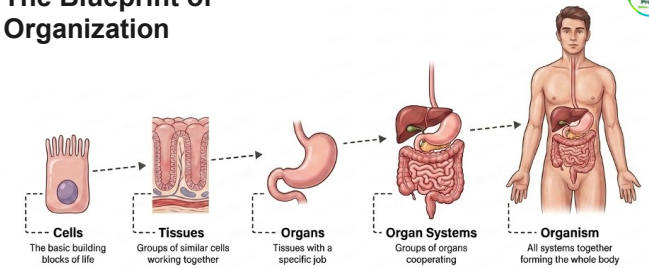
Physiology studies how body parts function and work together

Example: Anatomy describes **WHAT** the heart looks like; physiology explains **HOW** the heart pumps blood

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

4

The Blueprint of Organization



Small, specialized units continuously stack together to build the complete organism

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

5

The Digestive System

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

6

Digestion starts in your stomach

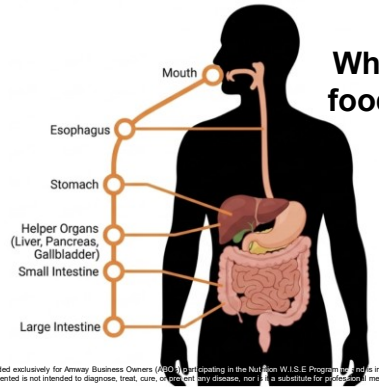
TRUE or FALSE?



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

7

Where does the food we eat go?

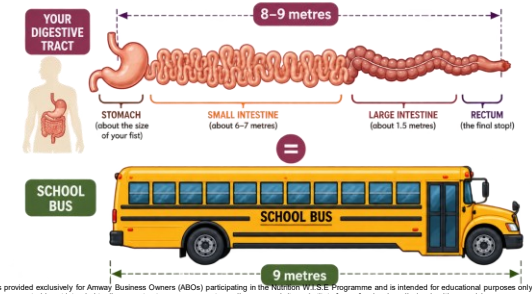


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

8

HOW LONG IS YOUR DIGESTIVE TRACT?

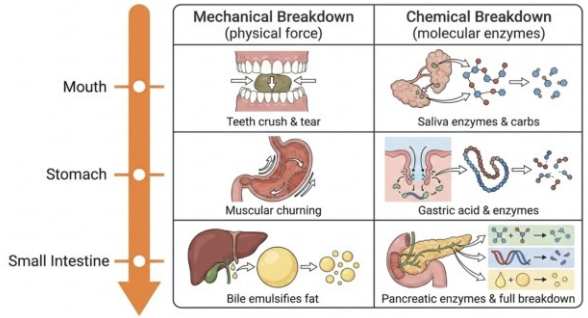
If you stretched it out, it's about 8–9 metres long – about the length of a schoot bus!



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

9

Mechanical vs Chemical Breakdown

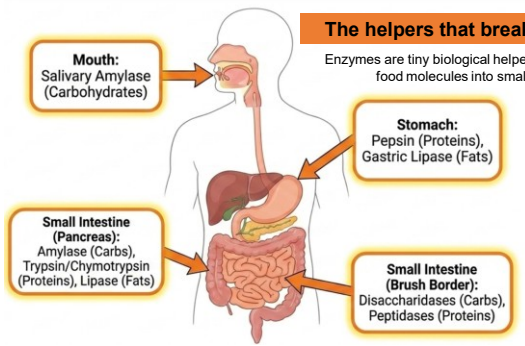


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

10

The helpers that break food down

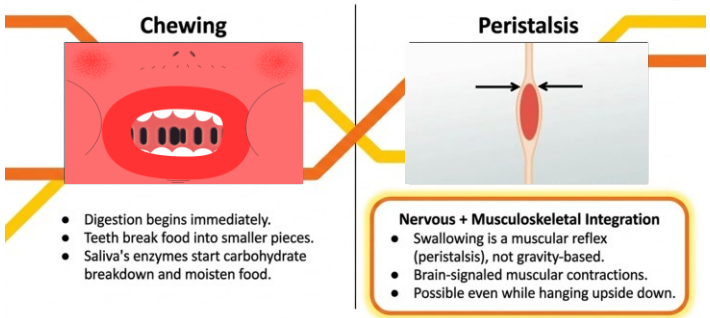
Enzymes are tiny biological helpers that break large food molecules into smaller pieces



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

11

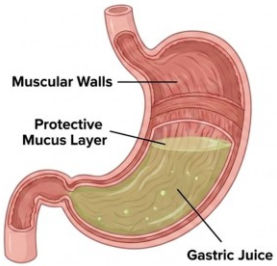
Station 1: Ingestion and the Swallowing Reflex



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

12

Station 2: The Stomach – Storage and Breakdown



Mechanical Function (Vat):

- Churns food (2-4 hrs).
- Converts to chyme.

Chemical Function & Defense (Fortress):

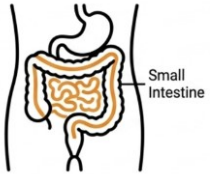
- Acidic juice kills pathogens.
- Mucus protects wall.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

13

Station 3: The Small Intestine – Where Digestion Happens

Why is the SMALL intestine the ***longest*** part of our digestive tract?

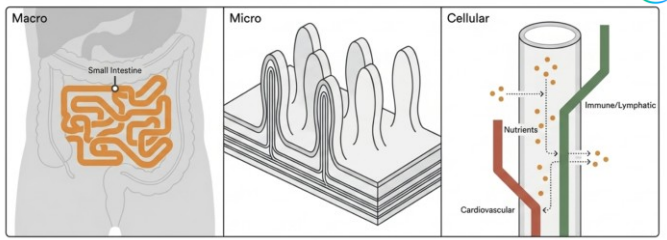




- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

14

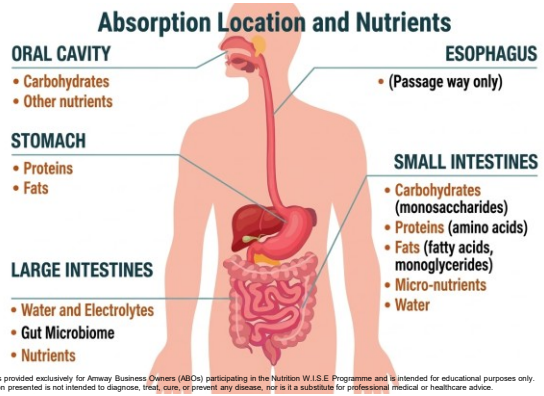
Station 3: The Small Intestine – Where Digestion Happens



The Interface: The small intestine is where digestion officially becomes fuel. The villi massively increase the surface area, allowing digested nutrients to finally cross the physical barrier of the digestive tract and enter the bloodstream.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

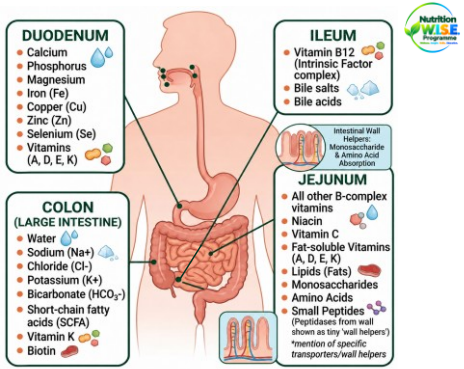
15



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

16

Nutrient absorption



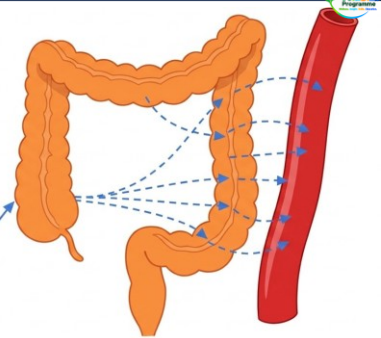
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

17

Station 4: Large Intestine – The Final Pass

The Final Pass:
- Receives leftover material after nutrient absorption.
- Absorbs water and remaining minerals.
- Turns waste into solid form for elimination.

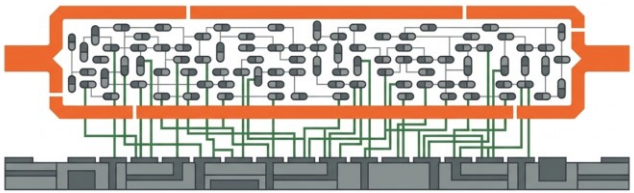
Renal System Integration:
- Intestine pumps reclaimed water into blood.
- Kidneys (Renal System) filter the blood.
- Balances salt and water levels; excess becomes urine.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

18

The Gut Microbiome



The Gut-Immune Connection: The large intestine is home to trillions of helpful bacteria. They are deeply integrated into our physiology across three pillars:

1. Digestion:
- Break down complex dietary fibers.
- Human enzymes cannot process these.

2. Synthesis:
- Actively produce essential vitamins.

3. Immunity:
- Crowd out harmful pathogens.
- Send chemical signals.
- Train and support adaptive immune system (T-Cells and B-Cells).

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

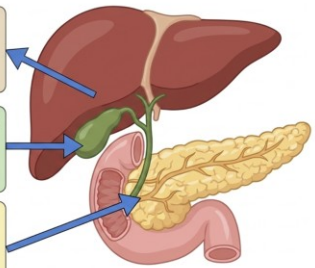
19

The Helper Organs

The Liver (The Multi-Tool)
- Produces bile for fat digestion
- Filters toxins from the blood
- Stores nutrients and energy

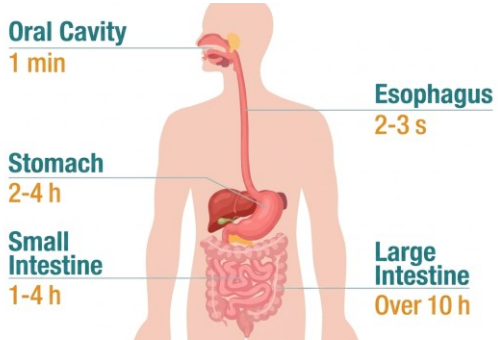
The Gallbladder (The Storage Unit)
- Stores and concentrates bile
- Releases bile into the small intestine after a meal

The Pancreas (The Engine)
- Produces enzymes to digest carbs, proteins, and fats
- Makes hormones like insulin to control blood sugar



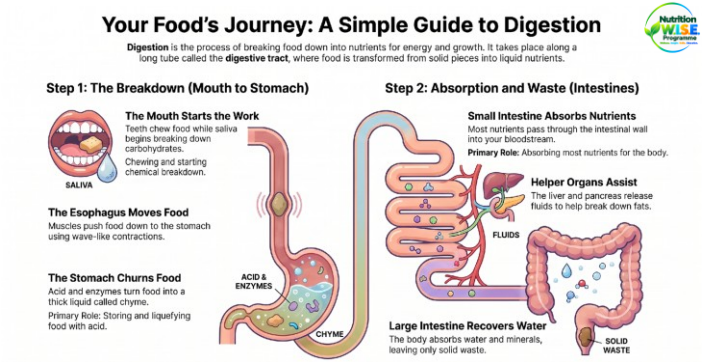
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

20



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

21



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

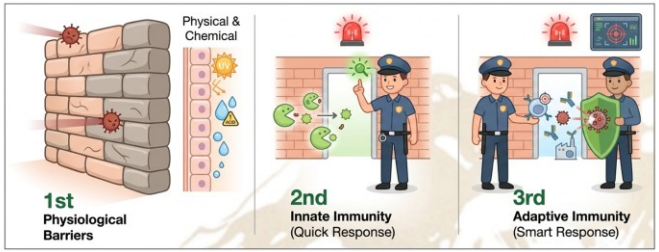
22

The Immune System

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

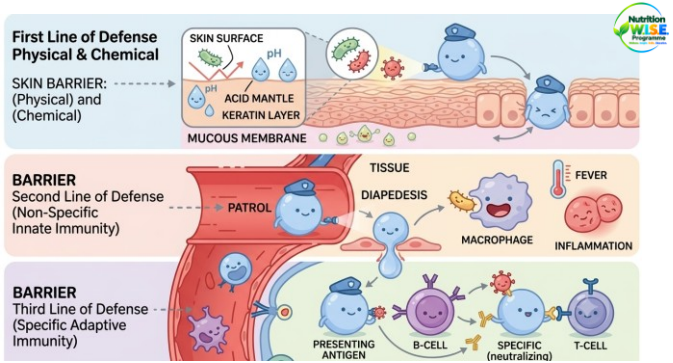
23

Our immune system operates as a three-tiered fortress



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

24



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

25

FIRST LINE OF DEFENSE

Physical and chemical barriers stop invaders at the gate

Skin: The body's largest organ. Features an acidic surface and natural oils that discourage germ growth.

Mucus & Cilia: Mucus traps invaders; tiny hair-like cilia sweep them away from lungs and throat.

Stomach Acid: High acidity kills many germs swallowed with food.

Tears & Saliva: Contain natural germ-fighting substances to wash away threats.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

26

SECOND LINE OF DEFENSE

The **innate immune system** is your **fast, general** responder

Always on Patrol: White blood cells travel continuously through the bloodstream alting acting as an early alert system.

General Defense: Responds the exact same way to most invaders without needing to "learn" who they are.

Minutes

Hours

Rapid Deployment: Activates within minutes to hours of detecting a threat.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

27

SECOND LINE OF DEFENSE

First responders use extreme tactics to clear local threats

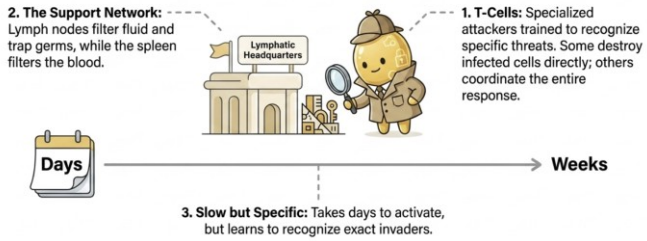
Phagocytosis: Swallowing and digesting invading germs

Inflammation & Fever: Redness, swelling, heat and pain. Higher body temperatures make it harder for germs to survive.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

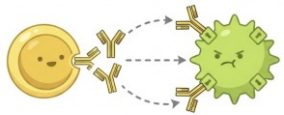
28

The **adaptive immune system** is a highly **SPECIFIC** intelligence network



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

B-cells craft custom “weapons” and build lifelong memory



Antibodies: Proteins that lock onto *specific* germs to mark them for destruction. Every antibody is **unique** to a particular invader.



Immune memory: After fighting an infection, some cells stay as *memory cells*. If the same germ returns, the response is **faster** and **stronger**.

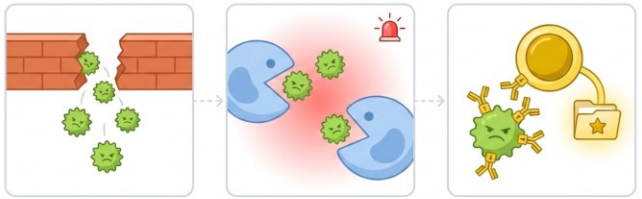
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

Comparing the two internal defense systems

Innate Immunity	Adaptive Immunity
Fast: Minutes to Hours	**Slow**: Days to Weeks
General: treats all threats equally	**Specific**: locks onto exact germs
No memory	Remembers past infections
Phagocytes & White Blood Cells	T Cells, B Cells & Antibodies

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

How the three lines work together when you get a paper cut



1. **Barrier Broken**: Germs enter the body.
2. **Innate Rush**: White blood cells cause red, swollen inflammation to fight early.
3. **Adaptive Cleanup**: T and B cells join to finish the fight and create memory cells.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

Vaccines build lasting memory without the danger of real illness

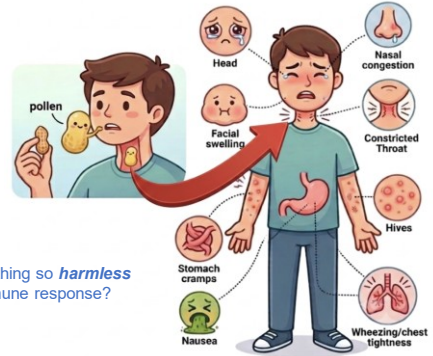


Vaccines introduce a safe, weakened, or partial version of a germ. This acts as a training exercise for the adaptive immune system.

The body builds **memory cells** based on the "**Wanted Poster**," ready to respond instantly and powerfully if exposed to the actual, dangerous germ later.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

33



How does something so ***harmless*** trigger an immune response?

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

34

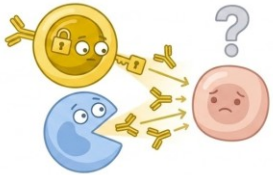
When the defense network misfires

The False Alarm



The immune system **overreacts** to a normally harmless substance (like pollen or food). It fights a fake threat, causing symptoms from mild sneezing to severe reactions.

Friendly Fire



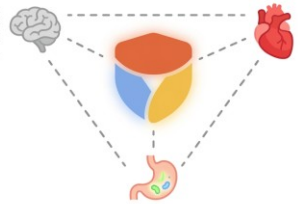
The immune system **mistakenly targets and attacks** the body's own **healthy cells** and organs. This requires diagnosis and management by a healthcare professional.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

35

Your immune system relies on the rest of your body to function

Nervous & Endocrine Systems: Communicate with immune cells to manage stress and illness responses.



Cardiovascular System: Transports and circulates immune cells through the bloodstream.

Digestive System: Helpful gut bacteria play a major role in regulating immune function.

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

36

Other Human Bodily System

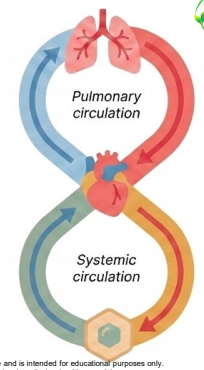
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

37

Profile of the Circulatory System

Function
Transports blood, oxygen, nutrients, and waste throughout the body.

Key Components
Heart, blood vessels (arteries, veins, capillaries), and blood.



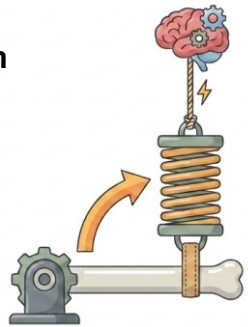
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

38

Profile of the Musculoskeletal System

System Goal
Provides form, support, stability, and movement to the body.

Key Components
Bones, Muscles, Cartilage, Tendons, Ligaments, Joints. These work together for movement and structural support.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

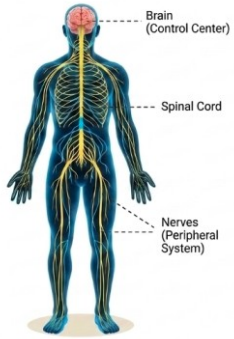
39

Profile of the Nervous System

System Goal
The body's rapid-response communication and control network.

Key Responsibilities
Controls voluntary movement, senses, thoughts, and automatic reflexes.

Key Components
- **Brain & Spinal Cord (CNS):** Command center for information processing.
- **Nerves (PNS):** High-speed electrical signal transmission between brain/cord and body.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

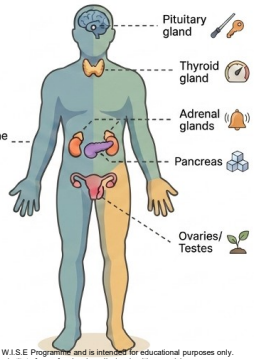
40

Profile of the Endocrine System

System Goal
The body's chemical messenger system.

Key Components
- Pituitary gland
- Thyroid gland
- Adrenal glands
- Pancreas
- Ovaries/Testes

Mechanism
Secrete hormones into the bloodstream to control long-term processes.



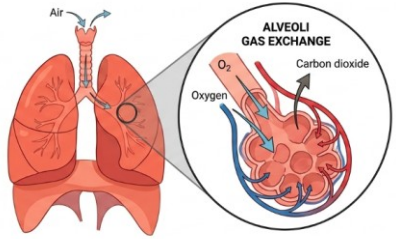
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

41

Profile of the Respiratory System

System Goal
Gas exchange: bringing in oxygen and removing carbon dioxide.

Key Components
Nose/Mouth, Trachea, Lungs, and Alveoli (where gas exchange occurs).



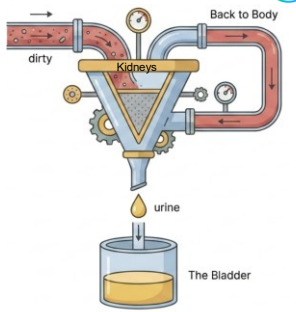
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

42

Profile of the Urinary System

System Goal
Filtering waste from blood and regulating fluid balance.

Key Components
Kidneys, Ureters, Bladder, and Urethra.



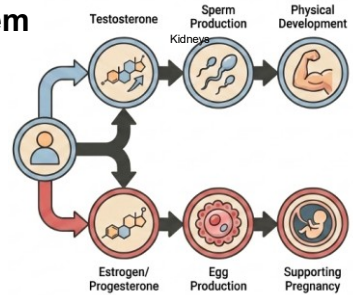
- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

43

Profile of the Reproductive System

System Goal
Produce offspring, regulate sexual development and hormones.

Key Components
Male: Testes (produce sperm & testosterone), ducts, glands, penis.
Female: Ovaries (produce eggs & hormones), fallopian tubes, uterus (nurtures fetus), vagina, mammary glands.

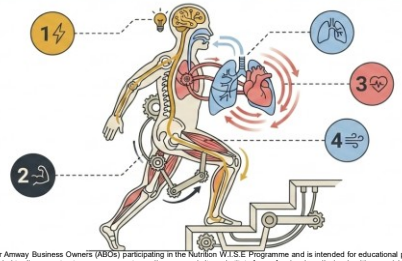


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

44

The Team in Action

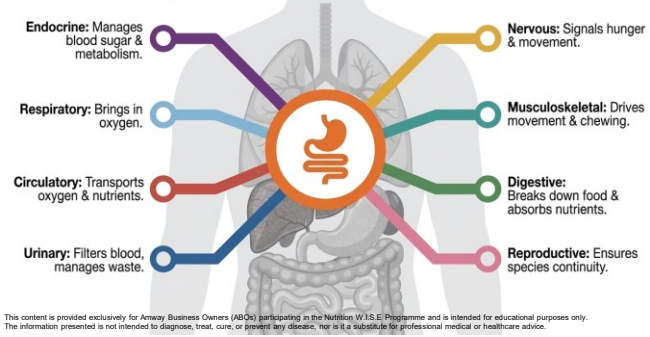
No single body system operates in isolation. A simple task like walking up a flight of stairs requires the Nervous system to send the signal, the Musculoskeletal system to pull the levers, the Cardiovascular system to deliver extra fuel, and the Respiratory system to pull in the necessary oxygen. Total system interdependence.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

45

The Blueprint: Eight Systems, One Mission



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

46

INTEGRATION

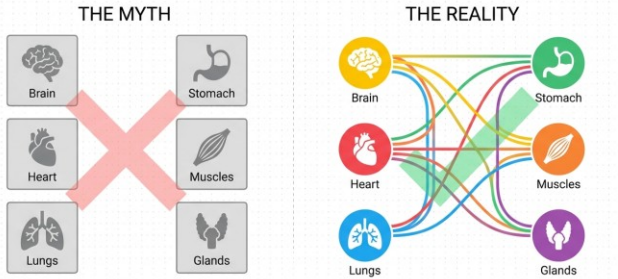
How does everything work together?

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

47

We often think of organs in isolation, but they never act alone.

Every action—running, eating, or even feeling startled—calls on multiple systems at once. Coordination, not isolation, is what keeps the body running smoothly.

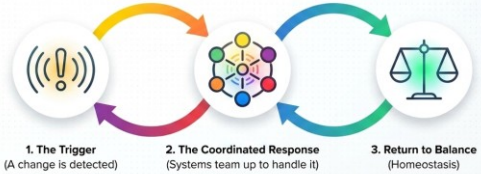


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

48

Every everyday action follows a predictable biological rhythm.

The body senses a change, mounts a coordinated response to manage it, and then works just as hard to return to a stable, internal environment known as homeostasis.



1. The Trigger
(A change is detected)

2. The Coordinated Response
(Systems team up to handle it)

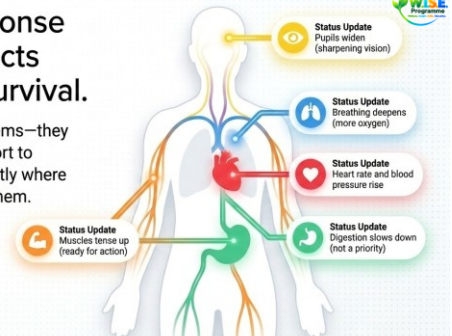
3. Return to Balance
(Homeostasis)

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

49

The stress response precisely redirects resources for survival.

These aren't separate problems—they are a highly coordinated effort to deliver oxygen and fuel exactly where your body expects to need them.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

50

Even resting after a meal requires an active, coordinated sequence.

A routine dinner calls on a highly synchronized team. The digestive system extracts nutrients, the cardiovascular system carries them away, and the endocrine system manages their storage.



1. Break Down
(Process food into nutrients)

2. Transport
(Increase blood flow to digestive tract)

3. Store
(Release hormones to store energy)

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

51

Occasional stress is healthy, but constant strain disrupts balance.

Occasional exertion is a normal part of life. But ongoing, frequent strain makes it harder for your systems to navigate back to their baseline balance.

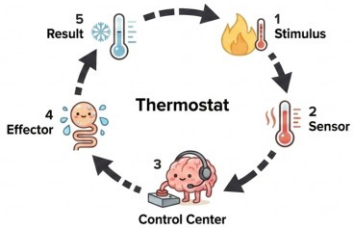


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

52

Homeostasis – What is it?

The body's automatic ability to maintain a stable internal environment despite external changes.

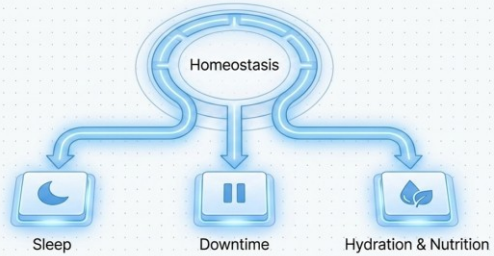


- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

53

Recovery provides the environment needed to recalibrate.

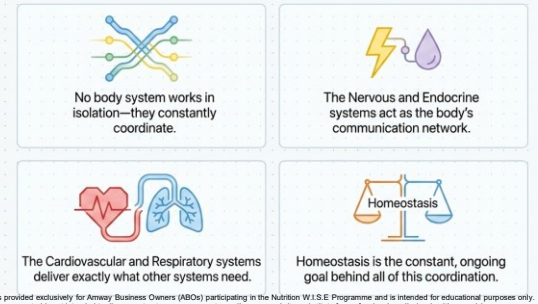
This is why rest, sleep, and recovery time are just as biologically important as activity.
They allow your interconnected systems to repair and prepare for the next trigger.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

54

The four principles of human physiology.



- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

55

SUMMARY

Key Takeaways

- **Your body is organized in levels** — cells → tissues → organs → systems → you
- **Digestion fuels everything** — turning food into energy & building blocks
- **Immunity works in three layers** — barriers, innate, and adaptive defense
- **Eight systems, one mission** — none of them works alone
- **Homeostasis ties it all together** — the constant goal behind every system

- This content is provided exclusively for Amway Business Owners (ABOs) participating in the Nutrition W.I.S.E Programme and is intended for educational purposes only.
- The information presented is not intended to diagnose, treat, cure, or prevent any disease, nor is it a substitute for professional medical or healthcare advice.

56